

Vibe-IC — ALL Steps, by Stage, in Order (v2.2.0)

One ordered list of every flow step, grouped by stage, with a single continuous number **1 → 33 starting at Stage 1 (Spec-to-RTL)**. Phase 1 doc-generation is the lettered pre-flow (**D1–D5**, not in the 1→33 count). Two parallel tracks — Analog (A1–A9) and Mixed-signal (M1–M4) — follow the main flow. Source of truth = the runners; finer code-level markers are auto-generated in `FLOW_STEPS_GENERATED.md`.

Stage map: **D** Spec/Docs (Phase 1) · **1** RTL+Verify · **2** Synthesis+DFT · **3** Physical+Sign-off · **4** Output+Tapeout.

Phase 1 (pre-flow) — Spec & Documents · D1–D5 (not in the 1→33 count)

#	Step	Tool / How
D1	Ingest & text extraction (prompt or vendor docs → input_doc/)	phase1_doc_one_shot_runner
D2	Generate L1–L13 core design-layer docs	deterministic extractors
D3	Generate L14–L23 protocol / timing / skeleton docs	overlay extractors
D4	Protocol-class synthesis dispatch (81 classes)	is_<proto> + <proto>_synth
D5	Coverage / parity report	phase1 parity report

Stage 1 — RTL Generation & Verification

#	Step	Tool / How
1	Spec-to-RTL (author RTL from L-docs)	spec-to-rtl skill
2	Lint	eda_lint + hygiene gates
3	CDC / RDC check	cdc-check
4	Simulation	testbench-gen + eda_simulate
5	Formal verification	formal-verify + assertion-gen
6	FPGA early prototype	eda_fpga_compile / eda_fpga_program → .sof

Stage 2 — Synthesis & DFT

#	Step	Tool / How
7	Constraint setup	constraint-gen → *.sdc
8	SDC validation	SDC lint
9	Synthesis (Yosys)	eda_synth + synth-doctor
10	Pre-layout STA	eda_sta_mcorner (SS/TT/FF)
11	DFT insertion (scan + ATPG)	dft-insert + atpg + eda_dft
12	Post-DFT optimization	resynth / buffering
13	Equivalence check (LEC)	equivalence-check + Yosys equiv

Stage 3 — Physical Design & Sign-off

#	Step	Tool / How
14	Floorplan + PDN	eda_pnr (init)
15	Clock planning	clock-planning skill
16	Placement (global + detailed)	eda_pnr
17	CTS	eda_pnr enable_cts=true

#	Step	Tool / How
18	Post-CTS hold fixing	repair_timing -hold
19	Routing (global + detailed)	eda_pnr enable_detailed_route=true
20	Parasitic extraction → SPEF	OpenRCX extract_parasitics
21	Post-route STA (MMMC)	eda_sta_mcorner
22	IR drop	OpenROAD PSM analyze_power_grid
23	EM check	PSM -enable_em
24	Antenna check	OpenROAD check_antennas
25	Signal integrity (crosstalk)	SPEF coupling-cap screen
26	Post-layout gate-level sim (+SDF)	eda_simulate
27	Physical verification (DRC / LVS / ERC + PERC-equivalent)	KLayout DRC + LVS sign-off chain + Magic ERC + perc_equivalent
28	ECO repair loop	eco-plan

Stage 4 — Output & Tapeout

#	Step	Tool / How
29	Power analysis	pre + post layout
30	Metal fill (density fill)	OpenROAD filler_placement
31	Tapeout checklist	tapeout-checklist + signoff_audit
32	GDSII output	eda_gds + def2gds
33	FPGA final sign-off	recompile + on-board test + attestation

Parallel track — Analog A1–A9 (`analog_one_shot_runner.py`, runs alongside Stages 1–3)

#	Step	Tool / How
A1	spec_extract	→ A1_spec.json
A2	topology_select	analog-topology-select → A2_topology.json
A3	netlist_gen	→ <block>.sp
A4	corner_sweep	ams-sim → A4_corners.json
A5	layout (Magic)	→ A5_layout.json
A6	per-block physical verification (DRC + LVS)	analog_a6_block_pv_check
A7	post-layout resim	→ A7_postsim.json
A8	hardmacro_gen	→ {.lef,.lib,.gds,.v} (feeds Stage 3)
A9	hw_verify (HIL) / co- sim	→ A9_hw_verify.json

Parallel track — Mixed-signal M1–M4 (mixed- signal-cosim skill, no dedicated runner)

#	Step	Tool / How
M1	top merge	mixed_signal_m1_top_merge_check.py
M2	co-sim setup	mixed-signal-cosim skill
M3	co-sim run	mixed-signal-cosim skill
M4	integration verify	mixed-signal-cosim skill

Totals

33 sequential steps — Stage 1: 1–6 · Stage 2: 7–13 · Stage 3: 14–28 · Stage 4: 29–33 — preceded by the lettered **Phase 1 pre-flow (D1–D5)**, plus **9 Analog (A1–A9)** and **4 Mixed-signal (M1–M4)** parallel steps.

Pre-flight: P0 (mcp_server_health_check, eda_doctor). Orchestrator `vibe_ic_one_shot_runner.py` runs Phase 1 → Phase 2 → Analog → Phase 3.

Detail beyond this summary: the runner's live code-level markers in `FLOW_STEPS_GENERATED.md`; the LVS sign-off chain + PERC-equivalent coverage in `PERC_SIGNOFF_MEMO.md`. 繁中版見 `ALL_STEPS_v2.2.0.zh-TW.md`.